

Traffic Management System for Real-Time Optimization of Light Timings on Congested Routes.

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Abstract

- AI-powered traffic control using real-time camera feeds.
- Uses **OpenCV** and **YOLOv8** for vehicle detection and counting.
- Dynamically adjusts traffic light timings based on congestion.
- Built on **Django** for web-based monitoring and control.
- Aims to reduce congestion and improve road efficiency.

Introduction

- Urban traffic congestion causes delays, pollution, and safety issues.
- Fixed-timing traffic lights fail to handle dynamic traffic conditions.
- Need for an intelligent and adaptive traffic management system.
- Proposed **AI-powered system** to optimize traffic flow in real-time.
- Uses **OpenCV** and **YOLOv8** for vehicle detection and classification.
- Goal: reduce congestion, improve safety, and enhance traffic efficiency.

Problem Objectives

- **Dynamic Timing Implementation:** Adjust light durations based on real-time traffic density.
- **Real-Time Density Analysis:** Use **OpenCV** and **YOLOv8** for accurate vehicle detection and congestion measurement.
- **Traffic Flow Optimization:** Reduce average wait time and improve road safety and fuel efficiency.
- **Scalable Platform Development:** Create a **Django-based web system** for centralized monitoring and management.

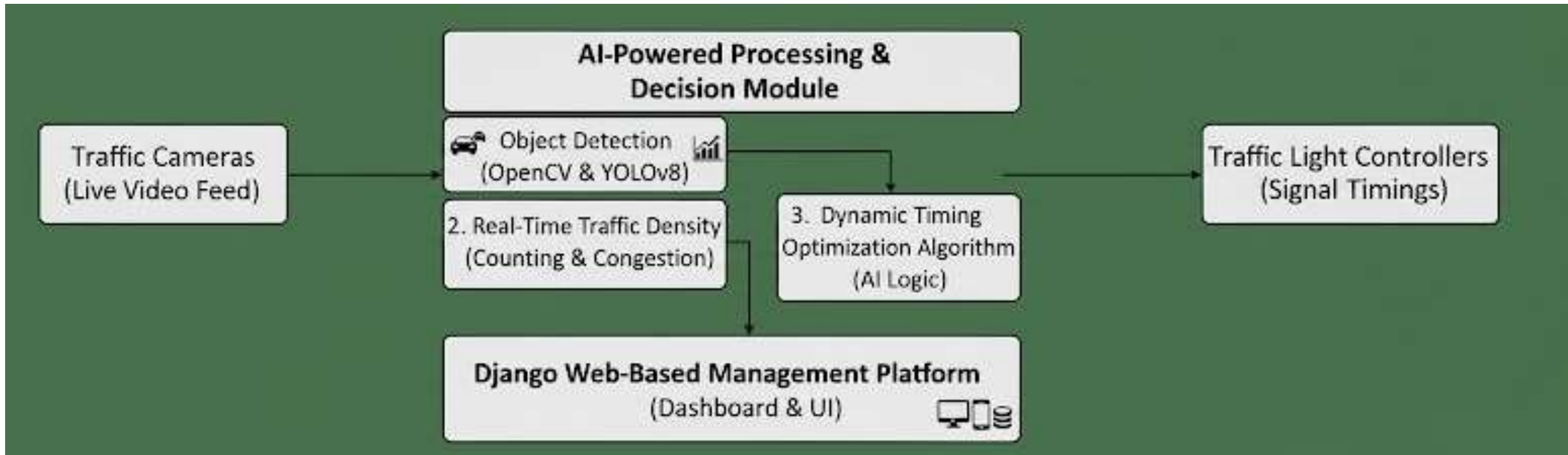
Problem Statement

- Conventional traffic systems lack adaptability and efficiency.
- **Fixed-time signals** fail to handle real-time traffic variations.
- Causes **increased congestion**, long waiting times, and delays.
- Leads to **higher pollution** and **reduced road safety**.
- Current systems lack **scalability** and real-time intelligence.

Literature Survey/ Related Works

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Proposed Model with Architecture Diagram



Modules Description

I. Data Acquisition Layer (Input)

Module	Purpose	Technology	Description
Traffic Camera Feed	Captures real-time video input of the intersection approaches.	IP Cameras (Assumed), Video Streaming Protocol	Feeds continuous video frames into the system's processing pipeline, acting as the primary data source for traffic conditions.

II. AI Processing & Decision Layer (The Core 70% - 30%)

This layer is where the "smart" decisions are made, forming the foundation of your project's technical complexity.

Module	Purpose	Technology	Description
1. Object Detection Module	Identifies and classifies every vehicle (car, bus, truck, motorcycle) in the video frames.	OpenCV (Video Processing), YOLOv8 (Object Detection Model)	Uses the pre-trained YOLOv8 model for fast and accurate single-shot detection. This module outputs precise bounding box coordinates and class labels for all detected vehicles.
2. Real-Time Density Analysis	Calculates the congestion level and vehicle count for each lane/approach.	Python/NumPy	Processes the raw bounding box data from the detection module. It defines virtual " counting zones " within the frame and tracks vehicles entering and exiting to provide a real-time, accurate vehicle density count for each traffic approach (e.g., North, South, East, West).

3. Dynamic Timing Optimization Algorithm

Determines the optimal green light duration based on the current traffic density.

Heuristic/AI Logic (Rule-Based), Python

This is the **decision-making core (the remaining 30% of your project)**. It takes the density counts as input and applies a set of rules (e.g., "If North approach density is 3x greater than all others, grant a 45-second green light cycle") to

4. Traffic Light Control Interface

Executes the new signal timings calculated by the AI.

Microcontroller/Actuator Simulation (or API call in a real deployment)

Receives the dynamic timing data (e.g., green for 45s, yellow for 5s) and sends the corresponding signals to the physical traffic lights to change their state.

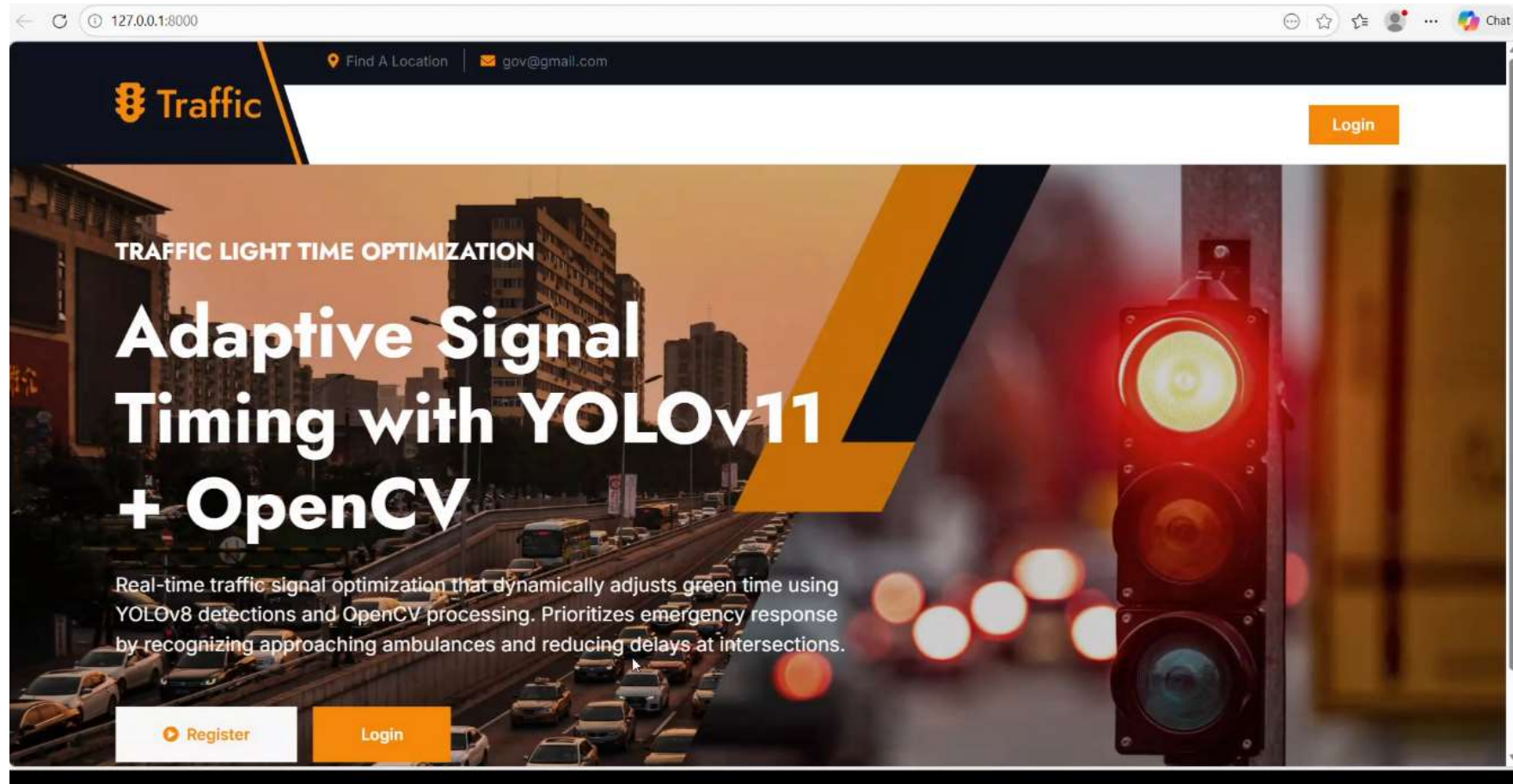
5. Django Web-Based Management Platform

Provides a central **dashboard** for monitoring and system control.

Django Framework (Python), HTML/CSS/JS

This interface allows traffic authorities to view the **real-time vehicle density**, track system performance metrics, and potentially adjust base parameters for the optimization algorithm, ensuring **scalability and usability**.

System Home Page



Video Upload Interface

 **Traffic Signal Optimization** 

Upload Video 1:

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Upload Video 2:

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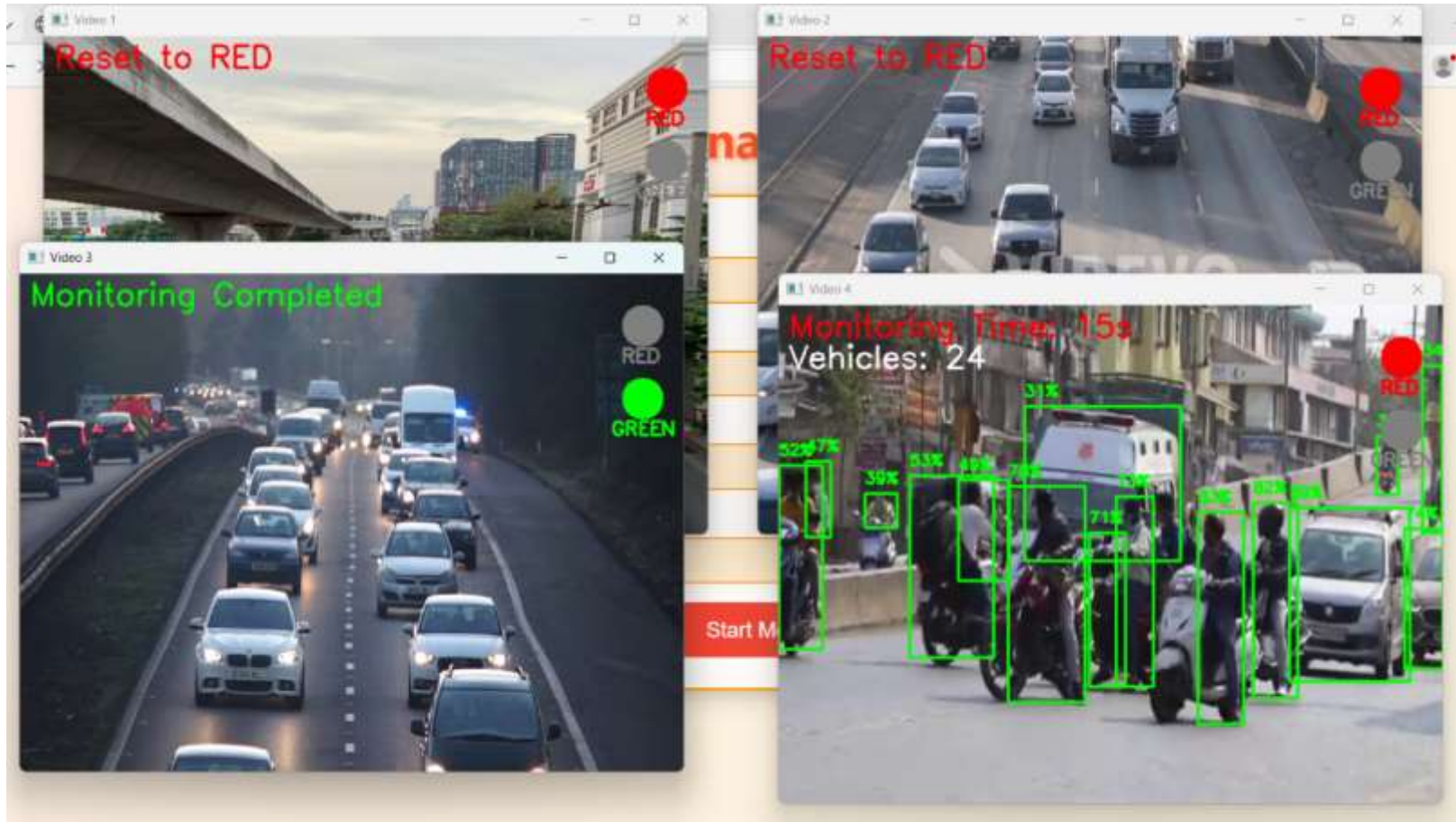
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Upload Video 4:

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Multi-Lane Traffic Monitoring



Traffic Analysis Results Graph



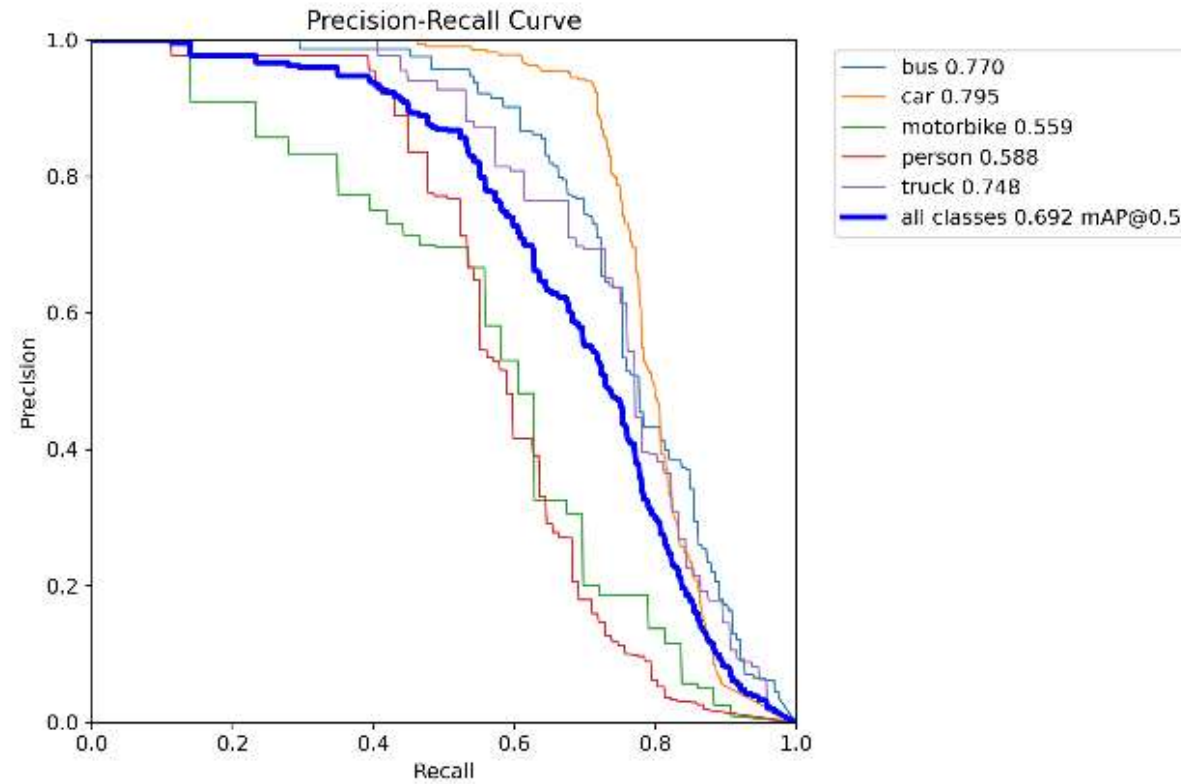
Signal Time Allocation Output

- ✓ Signal Time for Lane 1: 15 seconds
- ✓ Signal Time for Lane 2: 25 seconds
- ✓ Signal Time for Lane 3: 25 seconds
- ✓ Signal Time for Lane 4: 25 seconds

Emergency Vehicle Detection (Ambulance Priority)

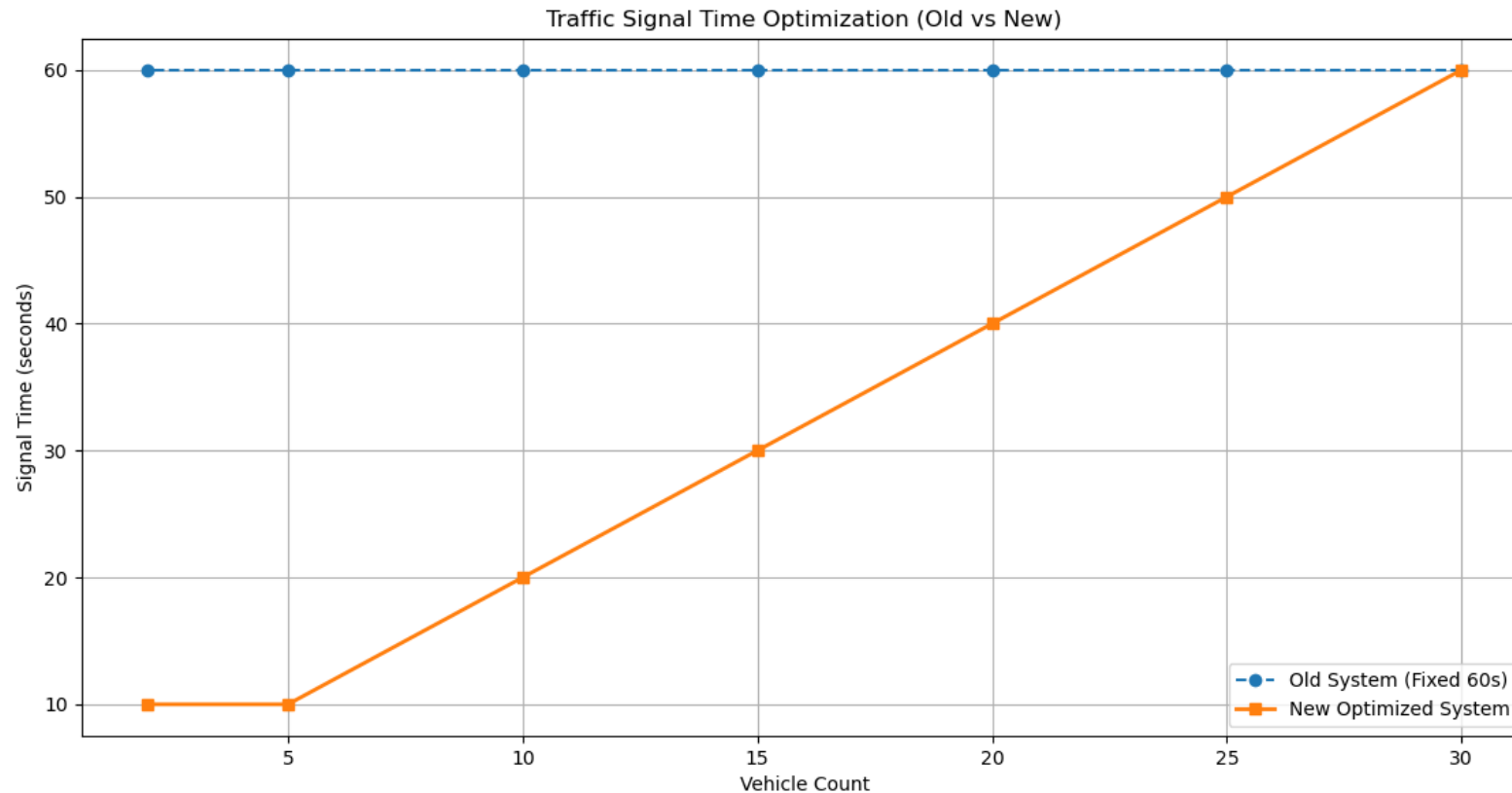


RESULTS



- **mAP @ IoU 0.5:** 0.692 (Overall accuracy)
- **Car Class AP:** 0.795 (Best performance for cars)

Fixed vs. Optimized Signal Timing



- Optimized signal gives more green time in high traffic
- Works effectively in heavy traffic conditions

CONCLUSION

- Real-time vehicle detection and classification
- Accurate traffic density analysis (lane-wise)
- Dynamic signal control based on vehicle count
- Reduces traffic congestion
- Cuts fuel consumption
- Supports smart city development

FUTURE SCOPE

- IoT sensor integration with cameras
- Multi-camera monitoring in real time
- Predictive traffic analysis using AI
- Smart city deployment at large scale
- Priority for emergency vehicles

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